Ops Runbook

The engineering-facing operations manual for PhysioGuidance — daily/weekly/monthly ops checks, incident response runbooks, and the specific commands to run when production needs attention.

PREPARED FOR
PhysioGuidance On-Call Engineer

DATE
2026-08-17

DEPLOYED COMMIT
d396b040

REGION
me-central2 (Dammam)

PHYSIOGUIDANCE
Riyadh · Saudi Arabia
me-central2 · GCP Cloud Run

Table of contents

1. Who this is for
2. Platform inventory
3. Daily ops checks (5 min)
4. Weekly ops checks (30 min)
5. Monthly ops checks (2 hours)
6. Incident runbooks
7. Common ops tasks
8. Access + tooling
9. On-call escalation
10. Component status matrix
11. Where to look next

---

1. Who this is for

This runbook is for the **engineer** on operational duty — someone with `gcloud` CLI access and IAM permissions to inspect and redeploy production. It complements platform-owner-guide.md, which is the same content shaped for the platform owner (Amna) working through the admin UI.

If you are the owner and want the button-click version, use platform-owner-guide.md instead.

---

2. Platform inventory

Five Cloud Run services run the platform in `me-central2`:

Service	Purpose	Deploy script
`physioguidance-api`	NestJS API (HTTP surface, crons)	`infra/backend/deploy.sh`
`physioguidance-web`	Next.js frontend	`infra/frontend/deploy.sh`
`physioguidance-cms`	Strapi CMS	`infra/strapi/deploy.sh`
`physioguidance-api-worker`	BullMQ worker (same image as api)	`infra/worker/deploy.sh`
`physioguidance-malware-scanner`	Eventarc-triggered ClamAV	`infra/malware-scanner/deploy.sh`

Supporting resources: Cloud SQL Postgres 15 instance `physioguidance-db-me`, Memorystore Redis `physioguidance-redis`, GCS buckets (`physioguidance-public`, `physioguidance-private`, `physioguidance-trainers`, `physioguidance-backups`, `physioguidance-quarantine`, `physioguidance-cms`), VPC connector `physio-vpc-connector-me`, and a global HTTPS load balancer fronting `physioguidance.org`, `www`, `api`, `cms`.

Cloud Run Jobs handle heavy ephemeral work: `physio-job-video-transcode`, `physio-job-captions-generate`, `physio-job-wcag-check`. Cloud Tasks queues drive event scheduling: `session-events`, `ticket-events`, `maintenance-events`, `points-events`, `backup-events`, `report-events`, `system-events`. Cloud Scheduler runs `reconciliation-daily-scan` (02:00 KSA) and `reconciliation-stale-reminders` (10:00 KSA) against the API service.

---

3. Daily ops checks (5 min)

Run every business morning. If everything is green, close the tab.

3.1 Backend health

```
curl -sS https://api.physioguidance.org/api/health/ready | jq .
```

Expect: `{"status":"ok","checks":{"database":{"ok":true,...},"redis":{"ok":true,...}}}`.

If the response is not `ok`, drop to §6.1.

3.2 All 5 services on latest revision at 100% traffic

```
gcloud run services list --region=me-central2 \
  --format="table(metadata.name,status.latestReadyRevisionName,status.traffic[0].percent)"
```

All five services show 100% traffic on their latest ready revision.

3.3 Sentry error rate baseline

Open the Sentry project and compare today's issue count with yesterday. If today is more than 2x yesterday, investigate the top new issue.

3.4 Cloud SQL running and last backup successful

```
gcloud sql instances describe physioguidance-db-me --format="value(state)"
gcloud sql backups list --instance=physioguidance-db-me --limit=1 \
  --format="value(status,windowStartTime)"
```

3.5 In-app Monitoring dashboard

`/dashboard/admin/monitoring` — confirm each widget shows data. Requires `monitoring.viewer` on the API runtime SA.

---

4. Weekly ops checks (30 min)

Run one weekday morning.

4.1 Cost review

GCP Console -> Billing -> Reports. Compare the last 7 days to the prior 7 days. If cost is more than 20% up, drop to §6.6.

4.2 Recent backup existence

```
gcloud sql backups list --instance=physioguidance-db-me --limit=10 \
  --format="table(id,windowStartTime,status)"
```

Expect 7 consecutive `SUCCESSFUL` entries.

4.3 Cloud Run Job failures last week

GCP Console -> Cloud Run -> Jobs (`physio-job-video-transcode` , `physio-job-captions-generate` , `physio-job-wcag-check`). Sort by last execution status. Investigate any failed executions in the logs.

4.4 Sentry issue triage

Sentry -> Issues -> Unresolved. Anything older than 7 days and unresolved is either fixed or explicitly marked ignored.

4.5 Bank-transfer receipts age

Admin -> Finance -> Bank Transfers -> pending. Any receipt older than 24 hours is flagged to finance.

---

5. Monthly ops checks (2 hours)

Run the first Monday of the month.

5.1 Secrets age review

```
gcloud secrets list --format="table(name,createTime)" --sort-by=createTime
```

Any secret older than 12 months is scheduled for rotation per secrets-rotation-plan.md.

5.2 IAM audit

```
gcloud projects get-iam-policy nelc-elearning-platform-479319 \
  --flatten="bindings[].members" \
  --filter="bindings.role:owner OR bindings.role:editor" \
  --format="table(bindings.role,bindings.members)"
```

Revoke any Owner or Editor binding that no longer belongs.

5.3 Log retention

```
gcloud logging buckets list --location=global \
  --format="table(bucketId,retentionDays)"
```

Both `_Default` and `_Required` show 400 days.

5.4 Cost forecast vs budget

GCP Console -> Billing -> Forecast. Next month's projection is compared against the 1,000 SAR budget alert threshold.

5.5 PDPL request log

Admin -> Compliance -> PDPL Requests. Any request past its 30-day response SLA escalates to the owner and legal counsel.

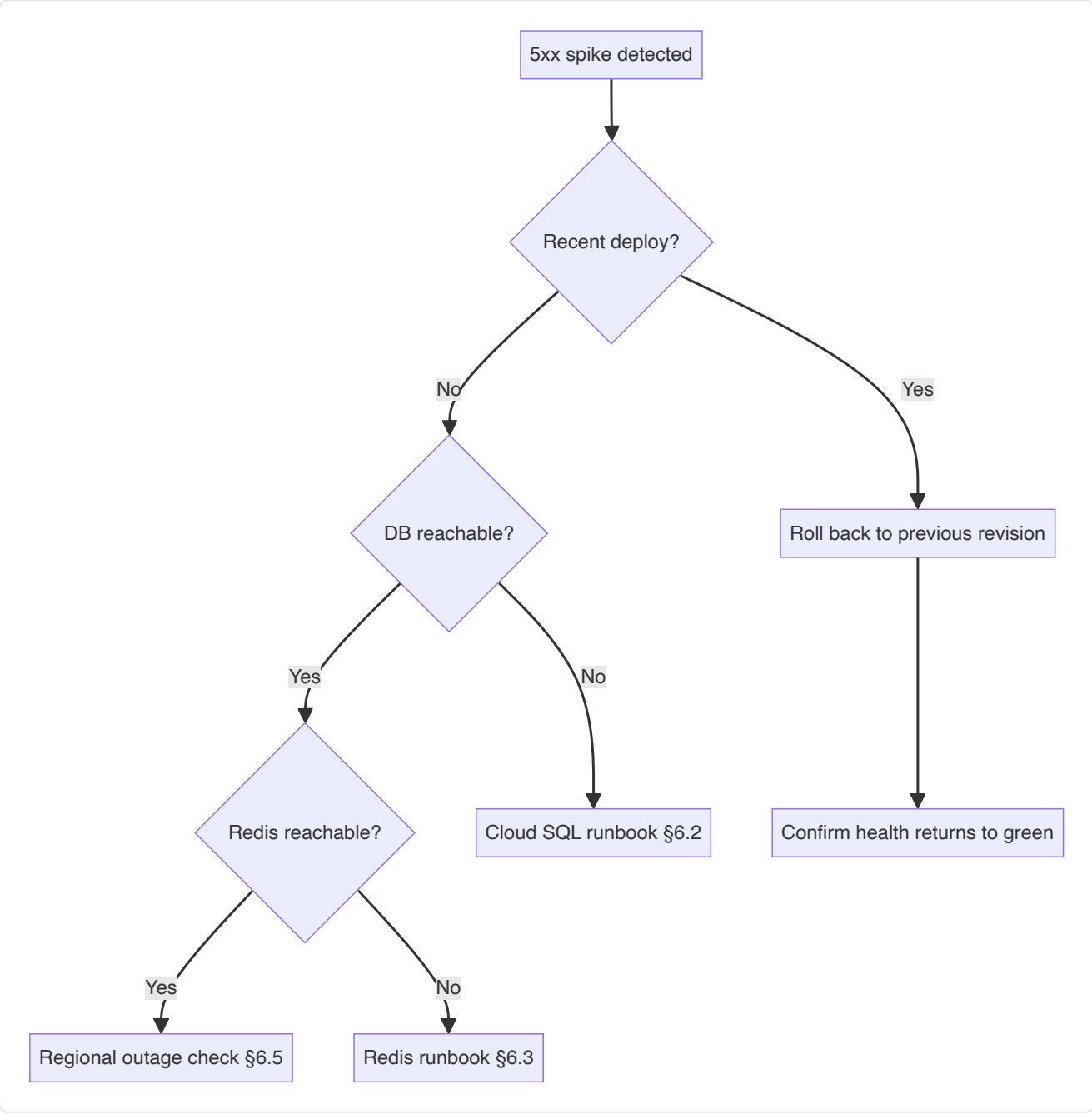
---

6. Incident runbooks

6.1 Site down / 5xx spike

Symptoms: users report an unreachable site, Sentry error rate is more than 10x baseline, uptime check fails.

Decision flow:



Step 1 — identify scope:

```
gcloud run services list --region=me-central2 \
  --format="table(metadata.name,status.latestReadyRevisionName)"
gcloud sql instances describe physioguidance-db-me --format="value(state)"
gcloud redis instances describe physioguidance-redis \
  --region=me-central2 --format="value(state)"
open https://status.cloud.google.com
```

Step 2 — recent deploy?

```
gcloud run revisions list --service=physioguidance-api \
  --region=me-central2 --limit=5
```

If the latest revision landed shortly before the incident, roll back per §7.1.

Step 3 — DB or Redis in trouble? Proceed to §6.2 or §6.3.

Step 4 — regional outage suspected? Proceed to §6.5.

6.2 Cloud SQL degraded

Zone outage (`me-central2-a` impaired, other zone healthy):

Full procedure in regional-failure-playbook.md §4.1. Short version:

```
gcloud sql instances clone physioguidance-db-me physioguidance-db-me-recovery \
  --preferred-zone=me-central2-b

printf '%s' "<new-connection-string>" | \
  gcloud secrets versions add DATABASE_URL --data-file=-

./infra/backend/deploy.sh
./infra/worker/deploy.sh
./infra/strapi/deploy.sh
```

Data corruption: the live `physioguidance` database is never restored in place. Two options:

1. **Admin UI restore-to-clone (preferred for a specific row):** Admin -> Settings -> System -> Backups -> Restore Clones -> Create From Backup. The API calls Cloud SQL Admin REST (`POST /databases` , `POST /import`) to reconstitute the chosen export into a new database on the same instance. The admin opens Cloud SQL Studio via the deep-link, extracts the row, and manually applies the fix to live. The sandbox auto-drops past `expiresAt` .
2. **`gcloud sql instances clone` at a PITR timestamp (fresh instance):** used when the whole database is corrupt. Full procedure in regional-failure-playbook.md §4.2.

6.3 Redis restart / cache loss

Redis is BASIC-tier without persistence. In-flight BullMQ jobs are lost on restart. Recovery is automatic per redis-persistence-risk.md:

- Payment reconciliation cron (every 5 min) re-fetches gateway status for any pending payment.

- `RecoverStuckRecordings` cron (hourly) re-enqueues WCAG jobs for recordings stuck longer than an hour.
- `xapi-flush` cron (every 2 min) re-sweeps failed and pending xAPI statements.

No manual action is typically required. Confirm Redis is back:

```
gcloud redis instances describe physioguidance-redis \
  --region=me-central2 --format="value(state)"
```

6.4 Payments stuck at "processing"

Symptoms: students report a payment stuck at processing, finance sees pending webhooks not resolving.

```
gcloud scheduler jobs describe reconciliation-daily-scan \
  --location=me-central2

gcloud logging read 'resource.type="cloud_run_revision"
  resource.labels.service_name="physioguidance-api"
  jsonPayload.event=~"reconciliation"' \
  --limit=20 --freshness=1h

open https://status.moyasar.com   # or tabby / tamara
```

If reconciliation is running but a specific payment is stuck for more than 15 minutes, use Admin -> Finance -> Payment detail -> Reconcile Now to force an immediate re-check.

If a full gateway is down, post an in-app maintenance banner (Admin -> Settings -> Maintenance) for that gateway only — other gateways continue to work.

6.5 Regional outage (me-central2)

Full procedure in regional-failure-playbook.md §4.4. Bottom line:

- **Wait for GCP** — PDPL and SDAIA prohibit cross-region failover.
- Post a maintenance banner via the domain registrar's holding page while the site is down.
- When the region recovers, run the post-outage verification checklist in §6 of that playbook.

6.6 Cost spike

Symptoms: budget alert email (50% / 90% / 100% / 120%).

```
gcloud alpha billing budgets list --billing-account=<account>
open https://console.cloud.google.com/billing
```

Common causes and fixes:

- Runaway Cloud Run Job (loop or excessive parallelism) — inspect Cloud Run Jobs execution history for unusually long runs.
- Malware scanner memory pressure — confirm the memory setting is 4 GiB per `infra/malware-scanner/deploy.sh`.
- Egress spike — check GCS bucket bandwidth for unusual download activity.
- Cloud SQL storage growth — auto-grow is off, so any growth is deliberate. Review the last change to the instance.

6.7 Suspected credential compromise

Full per-secret procedures in secrets-rotation-plan.md. Emergency short version:

1. Rotate the compromised secret immediately per the per-secret runbook.
2. If the compromise scope is unclear, rotate all third-party API keys as a precaution.
3. Force-reset admin passwords: Admin -> Users -> filter role=ADMIN -> bulk "Send Password Reset".
4. If PDPL Article 27 applies (PII may have been exposed), the owner and legal counsel escalate to SDAIA within 72 hours.

---

7. Common ops tasks

7.1 Roll back a service

Either the admin UI (Admin -> Settings -> System -> Backups -> Website Rollback, which calls `/admin/backups/website/rollback` and requires a typed domain confirmation) or gcloud:

```
gcloud run revisions list --service=physioguidance-api \
  --region=me-central2 --limit=5

gcloud run services update-traffic physioguidance-api --region=me-central2 \
  --to-revisions=physioguidance-api-00130-abc=100
```

Rollback takes about 10 seconds. In-flight requests on the current revision complete. Every deploy script re-pins traffic to `--to-latest`, so a subsequent deploy overrides a pinned rollback.

7.2 Force a redeploy without a code change

```
gh workflow run deploy-backend.yml
# or
./infra/backend/deploy.sh
```

Available workflows: `deploy-backend.yml`, `deploy-frontend.yml`, `deploy-strapi.yml`, `deploy-worker.yml`. The malware scanner deploys from a laptop via `./infra/malware-scanner/deploy.sh`.

7.3 Tail logs for a service in real time

```
gcloud run services logs tail physioguidance-api --region=me-central2
```

7.4 Query the last hour of errors

```
gcloud logging read 'resource.type="cloud_run_revision"
  resource.labels.service_name="physioguidance-api"
  severity>=ERROR' \
  --limit=50 --freshness=1h \
  --format="table(timestamp,severity,jsonPayload.message)"
```

7.5 Run a Cloud Run Job manually

```
gcloud run jobs execute physio-job-video-transcode \
  --region=me-central2 --wait
```

7.6 Trigger a Cloud Scheduler cron manually

```
gcloud scheduler jobs run reconciliation-daily-scan --location=me-central2
```

7.7 Rotate a secret

Full procedure per secret in secrets-rotation-plan.md §5. Short:

```
NEW_VALUE=$(openssl rand -base64 64 | tr -d '\n')
printf '%s' "$NEW_VALUE" | gcloud secrets versions add SECRET_NAME --data-file=-
# then redeploy consumers, verify, disable the previous version
```

The helper script `infra/scripts/manage-secrets.sh {list|get|set|delete} SECRET_NAME` wraps the same gcloud calls with prompts.

7.8 Apply a database migration

```
./infra/scripts/migrate-production.sh
```

Starts `cloud-sql-proxy` on `localhost:5433`, rewrites `DATABASE_URL` to point at the proxy, and runs `npm run db:migrate` in `backends/` — drizzle-kit uses its `__drizzle_migrations` tracking table. The one-shot runner at `backends/src/database/scripts/apply-single-migration.cjs` bypasses that tracking table and is reserved for local surgical work; production always goes through `migrate-production.sh`.

7.9 Create a manual backup snapshot

```
gcloud sql backups create --instance=physioguidance-db-me \
  --description="pre-migration snapshot 2026-08-17"
```

Or Admin -> Settings -> System -> Backups -> Create Snapshot (native Cloud SQL run) or Create Export (SQL dump to `physioguidance-backups` bucket, 30-day lifecycle, downloadable via a signed URL that expires after 30 minutes).

7.10 Restore drill (safe, throwaway)

Admin -> Settings -> System -> Backups -> Restore Clones -> Create From Backup. The service creates a new database on `physioguidance-db-me`, imports the chosen export, exposes a Cloud SQL Studio deep-link, and auto-drops the database past `expiresAt`. The live `physioguidance` database is never touched. Full procedure in regional-failure-playbook.md §4.2. Executed live on 2026-08-17 as part of the handover verification.

---

8. Access + tooling

8.1 IAM you need for common ops

Task	Required role
Read logs	`roles/logging.viewer`
Read monitoring metrics	`roles/monitoring.viewer`
Redeploy a service	`roles/run.developer` + `roles/artifactregistry.writer` + `roles/iam.serviceAccountUser`
Rotate a secret	`roles/secretmanager.admin` (per-secret preferred)
Manage Cloud SQL	`roles/cloudsql.admin` (grant temporarily; revoke after task)
Configure log retention	`roles/logging.configWriter`
Modify IAM policy	`roles/resourcemanager.projectIamAdmin`

8.2 Per-service runtime service accounts

Each Cloud Run service runs as its own least-privilege SA (verified in the deploy scripts):

Service	Runtime SA
`physioguidance-api`	`api-runtime@nelc-elearning-platform-479319.iam.gserviceaccount.com`
`physioguidance-web`	`web-runtime@nelc-elearning-platform-479319.iam.gserviceaccount.com`
`physioguidance-cms`	`strapi-runtime@nelc-elearning-platform-479319.iam.gserviceaccount.com`
`physioguidance-api-worker`	`worker-runtime@nelc-elearning-platform-479319.iam.gserviceaccount.com`
`physioguidance-malware-scanner`	`malware-runtime@nelc-elearning-platform-479319.iam.gserviceaccount.com`
Cloud Tasks dispatcher	`tasks-dispatcher@nelc-elearning-platform-479319.iam.gserviceaccount.com`
Cloud Scheduler (reconciliation)	`reconciliation-cron@nelc-elearning-platform-479319.iam.gserviceaccount.com`

8.3 Recommended local tooling

- `gcloud` CLI
- `docker`
- `jq` (JSON query)
- `pandoc` + Chrome (handover-doc regeneration, see `.tools/build-pdf.sh`)
- `pdftotext` (poppler-utils, reading legacy PDFs)
- `cloud-sql-proxy` (reaching Cloud SQL from your laptop)

---

9. On-call escalation

Situation	First responder	Escalate to
Site down more than 15 min after rollback	On-call engineer	Amna (owner)
Data loss suspected	On-call engineer	Amna + legal counsel
Sentry paging spike	On-call engineer	Self-resolve or PR fix
Payment gateway rejects live traffic	On-call engineer	Gateway vendor support
Compliance question (PDPL / NELC / SDAIA)	On-call engineer	Amna + legal counsel
GCP outage	On-call engineer	GCP support (paid tier)

10. Component status matrix

Using Amna's four-category vocabulary.

Item	Status	Notes
Health-check endpoint `/api/health/ready` returns real dep checks	Implemented	Verified 2026-08-17
Uptime checks for backend and frontend	Implemented	Every 5 minutes
Backup restore drill executed	Implemented	2026-08-17 restore-to-clone on `physioguidance-db-me`, verified, dropped
Rollback via Cloud Run traffic-shift	Implemented	Admin UI (`/admin/backups/website/rollback`) or gcloud, procedure §7.1
Migration workflow via `migrate-production.sh`	Implemented	Uses `cloud-sql-proxy` + drizzle-kit `__drizzle_migrations` tracking table
Sentry error monitoring (backend + frontend)	Implemented	Alerts route to Amna
In-app Monitoring dashboard	Implemented	6 widgets after `monitoring.viewer` grant
Long-term log retention	Implemented	400 days on `_Default` and `_Required`
Secret rotation runbooks	Implemented	secrets-rotation-plan.md — per-secret
Restore-to-clone architecture (no in-place restore)	Implemented	`RestoreCloneService` + `WebsiteRecoveryController`, see backup-system-decision.md

11. Where to look next

- **Owner-facing operations:** platform-owner-guide.md
- **Every deploy detail:** cicd-deployment.md
- **DR + regional recovery:** regional-failure-playbook.md
- **Every secret + rotation:** secrets-rotation-plan.md
- **Every env var:** environment-configuration-CONFIDENTIAL.md